

SYMBOLS



Field-mounted instrument (function letters / loop number)



On/off valve, electric actuator (XV)



Modulating control valve (TCV)



Pump



Magnetic flow meter (inline)



Signal to PLC (CP-1), shared display / control



Flow direction

LINES



Process piping (chilled water)



Secondary / connection piping



Electrical signal



Process connection / impulse line

IDENTIFICATION LETTERS (ANSI/ISA-5.1)

First letter (measured variable)	
F	Flow
L	Level
M	Moisture / humidity (user's choice)
P	Pressure
PD	Pressure, differential
T	Temperature
X	User's choice: on/off valve (XV), leak detection (XS)
Z	Position
Succeeding letters (function)	
T	Transmit
S	Switch
V	Valve
CV	Control valve (TCV)
SO / SC	Switch, open / closed (position proof)
SL	Switch, low

GENERAL NOTES

1. Tags: <letters>-<loop>. Loop series: 1xx chillers, 2xx primary pumps, 3xx secondary pumps, 4xx CRAHs, 5xx headers / decoupler / expansion tank, 6xx mode switching and outdoor.
2. Equipment points (start/stop, status, alarms, VFD speed) are listed in the I/O list, not shown as bubbles.
3. Fail positions are a system decision: on total loss of signal the chiller paths open and the economizer path closes, so the primary pumps always have a flow path.

ABBREVIATIONS

CHWS / CHWR	Chilled water supply / return
CW	Condenser water
CRAH	Computer room air handler
WSE	Waterside economizer
ET	Expansion tank
VFD	Variable frequency drive
FO / FC	Fail open / fail closed
HP / LP	High / low pressure tap
SA / RA	Supply / return air
PLC	Programmable logic controller (CP-1)
DN	Nominal diameter (mm)

NOT FOR CONSTRUCTION - PORTFOLIO PROJECT

REV	DATE	DESCRIPTION
A	2026-10-08	ISSUED FOR REVIEW

Data Center Chilled Water Plant - I&C Package	
P&ID LEGEND AND NOTES	
DWG DC-I-002	REV A
DATE 2026-10-08 SCALE NTS DRAWN S. Anandkumar	SHEET ANSI B